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Inventor(s)

Cheng; Chuang-Yuan et al.

ELECTRONIC DEVICE AND CONTROL METHOD THEREOF

Abstract

An electronic device includes a first body, a second body, a rotating module, and at least one light-emitting module. The rotating module is connected to the first body and the second body. The at least one light-emitting module is disposed on the rotating module. The second body is adapted to drive the rotating module to rotate relative to the first body along an axial direction to switch to an unfolding mode or a closing mode. In the closing mode, the at least one light-emitting module faces a rear end of the first body. In the unfolding mode, the at least one light-emitting module faces an upper surface of the first body.

Inventors: Cheng; Chuang-Yuan (Taipei City, TW), Huang; Chen-Yi (Taipei City, TW), Chien; Po-Yang (Taipei City, TW), Chen; Po-Jui (Taipei City, TW), Chen; Chun-Chieh (Taipei City, TW), Pai; Yun-Tung (Taipei City, TW), Chiang; Chih-Wen (Taipei City, TW)

Applicant: COMPAL ELECTRONICS, INC. (Taipei City, TW)

Family ID: 96881271

Assignee: COMPAL ELECTRONICS, INC. (Taipei City, TW)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the priority benefit of U.S. provisional application Ser. No. 63/561,234, filed on Mar. 4, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to an electronic device, and more particularly, to an electronic device with a light-emitting function and a control method thereof.

Description of Related Art

[0003] In an existing electronic device with light effects, a light-emitting module is usually installed at a specific position on a display body or a host body. If the light-emitting module is installed on a front side, a user may not show the light effects to others when using it. If the light-emitting module is installed on a back side or both sides, the user may not directly see the light effects.

[0004] In short, in the existing electronic device, an installation position of the light-emitting module is fixed, and the user may not adjust the position of the light-emitting module by himself, which results in limitations in a light-emitting function. If the light effects on the front, back, or both sides are to be achieved, multiple sets of light-emitting modules are required to be installed, which has disadvantages of taking up the space of the body and being expensive.

SUMMARY

[0005] The disclosure provides an electronic device that changes an orientation of a light-emitting module in an unfolding mode and a closing mode, thereby allowing a light to be transmitted to different positions of the electronic device.

[0006] An electronic device in the disclosure includes a first body, a second body, a rotating module, and at least one light-emitting module. The second body is adapted to display a frame. The rotating module is connected to the first body and the second body. The at least one light-emitting module is disposed on the rotating module. The second body is adapted to drive the rotating module to rotate relative to the first body along an axial direction to switch to an unfolding mode or a closing mode. In the closing mode, a light-emitting direction of the at least one light-emitting module faces a rear end of the first body. In the unfolding mode, the light-emitting direction of the at least one light-emitting module faces a front end of the first body relative to the rear end.

[0007] The disclosure provides a control method of an electronic device, which generates the corresponding light according to different conditions.

[0008] The control method of the electronic device in the disclosure includes the electronic device and a processing unit, which is coupled to the at least one light-emitting module and controls the at least one light-emitting module to generate a light.

[0009] Based on the above, in the electronic device of the disclosure, the light-emitting module is installed on the rotating module, so the illumination angle may be changed with the rotation of the rotating module. In the closing mode, the light-emitting module faces the rear end of the first body, allowing the light to propagate in a direction away from the user and display it to others. In the

unfolding mode, the light-emitting module faces the upper surface of the first body, allowing the light to propagate to the front and the two sides of the first body for display to the user. Furthermore, the electronic device of the disclosure does not require multiple light-emitting modules to be installed at multiple positions of the electronic device. Compared to the conventional technology, it may improve disadvantages of taking up the space of the body and being expensive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1A is a schematic partial side view of a closing mode of an electronic device according to an embodiment of the disclosure.

[0011] FIG. 1B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 1A.

[0012] FIG. 1C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 1B.

[0013] FIG. 1D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 1A.

[0014] FIG. 1E is a schematic block diagram of some elements of an electronic device according to an embodiment of the disclosure.

[0015] FIG. 1F is a flow chart of a control method of the electronic device in FIG. 1E.

[0016] FIG. 2 is a schematic perspective view of an unfolding mode of an electronic device according to another embodiment of the disclosure.

[0017] FIG. 3A is a schematic partial side view of a closing mode of an electronic device according to another embodiment of the disclosure.

[0018] FIG. 3B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 3A.

[0019] FIG. 3C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 3B.

[0020] FIG. 3D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 3A.

[0021] FIG. 4A is a schematic partial side view of a closing mode of an electronic device according to another embodiment of the disclosure.

[0022] FIG. 4B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 4A.

[0023] FIG. 4C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 4B.

[0024] FIG. 4D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 4A.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0025] FIG. 1A is a schematic partial side view of a closing mode of an electronic device according to an embodiment of the disclosure. FIG. 1B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 1A. FIG. 1C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 1B. FIG. 1D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 1A.

[0026] Referring to FIGS. 1A, 1B, and 1D, an electronic device **100** in this embodiment is formed by a logic body and a display body. The logic body is used to accommodate various electronic elements such as a central processing unit, a graphics chip, a hard drive, and a heat dissipation module, and has a protective function to prevent the electronic device **100** from being interfered by external electromagnetic signals or accumulation of dust during operation. The display body is

electrically coupled to the logic body, and is used to output an image frame.

[0027] The electronic device **100** in the disclosure includes a first body **110**, a second body **120**, a rotating module **130**, and at least one light-emitting module **140**.

[0028] The first body **110** is a logic body, and the second body **120** is a display body. The rotating module **130** is connected to the first body **110** and the second body **120**, so that the first body **110** and the second body **120** are suitable for relative unfolding and relative approaching. Furthermore, the second body **120** includes a screen **121**, and the screen **121** is used to display a frame P. The frame P includes a picture, an image, and a system operating interface.

[0029] Referring to FIGS. **1C** and **1D**, the rotating module **130** is a three-section structure and includes two positioning portions **131** and a rotating portion **132**. The two positioning portions **131** are fixed on a rear end RE of the first body **110**, and are spaced apart from each other to form an accommodating space. The rotating portion **132** is fixed on the second body **120**, and is pivotally connected to the two positioning portions **131**. The accommodating space is for the second body **120** to rotate.

[0030] Referring to FIGS. **1C** and **1D**, the at least one light-emitting module **140** is disposed on the rotating portion **132** of the rotating module **130**, and is located in the accommodating space. In addition, the at least one light-emitting module **140** is coupled to the first body **110** to receive an instruction. The light-emitting module **140** is formed by one or more light-emitting diodes, and a light color and a light-emitting frequency of the light-emitting module **140** are controlled by a processing unit **111** in the first body **110**. The at least one light-emitting module **140** has a light-emitting direction LD, which is away from an axial direction AD of the rotating module **130**. There is an included angle C between the light-emitting direction LD and the second body **120**, and the included angle C is between 0 degrees and 60 degrees.

[0031] This means that the light-emitting direction LD is parallel to an extension plane of the second body **120**, or the included angle C is between the light-emitting direction LD and the extension plane of the second body **120**.

[0032] Referring to FIGS. **1A** and **1B**, the second body **120** is adapted to drive the rotating module **130** to rotate relative to the first body **110** along the axial direction AD to switch to the unfolding mode or the closing mode. In addition, a light L of at least one light-emitting module **140** propagates toward the axial direction AD away from the rotating module **130**.

[0033] Referring to FIGS. **1A** and **1D**, in the closing mode, the light-emitting direction LD of the at least one light-emitting module **140** faces the rear end RE of the first body **110**. That is, the light L of the at least one light-emitting module **140** is adapted to directly or indirectly illuminated to the rear end RE of the first body **110**. Specifically, referring to FIG. **1A**, the light L of the at least one light-emitting module **140** is parallel to the first body **110**, or the included angle C is between the light L and the second body **120**. An angle range of the included angle C is between 0° and 60°.

[0034] Referring to FIGS. **1B** and **1C**, in the unfolding mode, the light-emitting direction LD of the at least one light-emitting module **140** faces a front end FE of the first body **110** relative to the rear end RE. That is, the light L of the at least one light-emitting module **140** is adapted to directly or indirectly illuminated to the front end FE and an upper surface TS of the first body **110**.

[0035] Referring to FIGS. **1A**, **1B**, and **1E**, the electronic device **100** includes a bar-shape cover **150**, a reflective member **160** and a frame sensor **170**.

[0036] The bar-shape cover **150** is disposed on the rear end RE of the first body **110**, and has a recess portion **151** and an extension portion **152**. The bar-shape cover **150** is formed by a translucent material. The recess portion **151** accommodates the rotating portion **132** of the rotating module **130** and at least one light-emitting module **140**. The extension portion **152** is connected to the recess portion **151**, and is away from the rear end RE of the first body **110**. The reflective member **160** is disposed in the first body **110** and aligned with the recess portion **151** and the extension portion **152**. That is, the reflective member **160** is located below the recess portion **151** and the extension portion **152**. The frame sensor **170** is disposed on the upper surface TS of the first

body **110** or the bar-shape cover **150**.

[0037] Referring to FIG. **1A**, in the closing mode, the light **L** of the at least one light-emitting module **140** passes through the recess portion **151** to be illuminated to a heat dissipation frame **180** at the rear end **RE**, and then the light **L** passes through the heat dissipation frame **180** to propagate to the outside world. The light **L** will be distributed at the rear end **RE** of the first body **110** for visual viewing by outsiders. Furthermore, the heat dissipation frame **180** may be a fin protective case.

[0038] Referring to FIGS. **1B** and **1C**, in the unfolding mode, there is an angle **A** between the second body **120** and the first body **110**. The light **L** of the at least one light-emitting module **140** passes through the recess portion **151** to be illuminated to the reflective member **160**, and the reflective member **160** reflects the light **L** to the extension portion **152**. In short, the at least one light-emitting module **140** rotates with the second body **120**, so that the at least one light-emitting module **140** faces the recess portion **151**. The light **L** thereof propagates to the recess portion **151**, the reflective member **160**, and the extension portion **152** in sequence, and finally the light **L** is distributed on a surface of the extension portion **152**, so as to increase a light-emitting range of the front end **FE** and two sides **SE** of the first body **110**. The frame sensor **170** detects a frame color of the screen **121** or a color of an ambient light source in both the unfolding mode and the closing mode.

[0039] Referring to FIGS. **1C** to **1E**, the light-emitting module **140** in this case reflects the frame color of the screen **121** of the second body **120** according to a sensing result of the frame sensor **170** to generate the light **L** of the corresponding color, which is applicable to the unfolding mode and the closing mode.

[0040] For example, when the electronic device **100** is not connected to an external screen **190**, the processing unit **111** of the first body **110** processes all area frames of the second body **120** to obtain a representative color and instruct the light-emitting module **140** to generate the corresponding light **L**. When the electronic device **100** is connected to the external screen **190**, the processing unit of the first body **110** processes all area frames of the external screen to obtain the representative color and instruct the light-emitting module **140** to generate the corresponding light **L**. In this way, the light-emitting module **140** may generate a color effect similar to that of a display frame.

[0041] Referring to FIGS. **1C**, **1E**, and **1F**, the following describes a control method of an electronic device in this embodiment. The processing unit **111** is coupled to and controls the at least one light-emitting module **140** to generate the light **L**. In step **S1**, a detection unit **112** is coupled to the processing unit **111**, and is used to detect whether the electronic device **100** is connected to the external screen **190**. In step **S2**, if yes, the processing unit **111** controls the external screen **190** to display the frame **P**. In step **S3**, if not, the processing unit **111** controls the second body **120** to display the frame **P**. Regardless of whether the electronic device **100** displays the frame **P** through the external screen **190** or the second body **120**, in step **S4**, the processing unit **111** processes some of areas or all of the areas of the frame **P** to obtain a representative color of some of the areas or all of the areas. In step **S5**, then, the processing unit **111** controls the light-emitting module **140** to generate the light **L** corresponding to the representative color according to the representative color. In addition, the frame sensor **170** is coupled to the processing unit **111**, and the frame sensor **170** is adapted to sense the color of the ambient light source. The processing unit **111** corrects the representative color according to the color of the ambient light source and controls the light **L** corresponding to the representative color.

[0042] Based on the above, the way in which the electronic device **100** in this case obtains a representative color of a game frame is related to whether it is connected to the external screen **190**. By changing a color of the light-emitting module **140**, the users or others may know a screen connection state of the electronic device in this case.

[0043] Referring to FIG. **2**, there are differences between an electronic device **100A** in this embodiment and the electronic device **100** shown in FIG. **1C**. The difference is that the electronic

device **100A** includes a frame sensor **170a**, which is disposed on a bar-shape cover **150a**, and the bar-shape cover **150a** in this embodiment is formed by a transparent material. The frame sensor **170a** detects a frame color of a second body **120a** or the color of the ambient light source in the unfolding mode.

[0044] When the electronic device **100A** is connected to the external screen, the processing unit processes all of the areas of the frame to obtain the representative color of the frame.

[0045] When the electronic device **100A** is not connected to the external screen, the processing unit processes some of the areas of the frame or the ambient light source to obtain the representative color of the frame.

[0046] Some of areas of a frame of the body **120a** may be preset or defined by a range indicated by the user.

[0047] If some of the areas are upper areas, an effect that the color of the light is the same or similar to a representative color of the upper area may be generated. If the representative color of the upper area is different from a representative color of a lower area, an effect of highlighting or gradation of a frame of the lower area may be generated. If some of the areas are the lower areas, an effect that the color of the light is the same or similar to the representative color of the lower area and the frame is extended may be generated.

[0048] When the frame sensor **170a** processes the ambient light source, color correction may be performed on a representative color of a frame of the second body **120a** according to the color of the ambient light source. Since the way in which the electronic device **100A** obtains the representative color is related to whether it is connected to the external screen, the color of the light is changed to allow the user or others to know a screen connection state of the electronic device **100A**.

[0049] FIG. 3A is a schematic partial side view of a closing mode of an electronic device according to another embodiment of the disclosure. FIG. 3B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 3A. FIG. 3C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 3B. FIG. 3D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 3A.

[0050] Referring to FIGS. 3A to 3D, there are differences between an electronic device **100B** in this embodiment and the electronic device **100** shown in FIG. 1D. The difference is that a rotating module **130b** of the electronic device **100B** has a five-section structure and includes two positioning portions **131b** and three rotating portions **132b**. The two positioning portions **131b** are fixed on the rear end RE of a first body **110b**, and are spaced apart from each other. The three rotating portions **132b** are fixed on a second body **120b**, and are pivotally connected to the two positioning portions **131b** respectively. That is, the middle rotating portion **132b** is pivotally connected to inner sides of the two positioning portions **131b**, and the rotating portions **132b** on two sides are pivotally connected to outer sides of the two positioning portions **131b** respectively.

[0051] Referring to FIGS. 3C and 3D, at least one light-emitting module **140b** includes three, which are respectively disposed on the three rotating portions **132b** of the rotating module **130b**.

[0052] The electronic device **100B** includes a light guiding member **160b**, which is disposed in the first body **110b** and aligned with an extension portion **152b** of a bar-shape cover **150b**. The bar-shape cover **150b** is formed by the translucent material.

[0053] Referring to FIGS. 3A and 3D, in the closing mode, the light L of each of the light-emitting modules **140b** passes through a recess portion **151b** to be illuminated to a heat dissipation frame at the rear end RE, and then the light L passes through the heat dissipation frame to propagate to the outside world. The light L will be distributed at the rear end RE of the first body **110b** for visual viewing by the outsiders.

[0054] Referring to FIGS. 3B and 3C, in the unfolding mode, the light L of each of the light-emitting modules **140b** passes through the extension portion **152b** to be illuminated to the light guiding member **160b**, and the light guiding member **160b** guides the light L to a full range of the

extension portion **152b**. In short, each of the light-emitting modules **140b** rotates with the second body **120b**, so that each of the light-emitting module **140b** faces the extension portion **152b**. The light L thereof propagates to the extension portion **152b** and the light guiding member **160b** in sequence, and finally the light guiding member **160b** guides the light L and distributes it on a surface of the extension portion **152b**, so as to increase the light-emitting range of the front end FE and the two sides SE of the first body **110b**.

[0055] Referring to FIG. 3C, when the electronic device **100B** is connected to the external screen, the processing unit processes some of areas of a frame of the second body **120b** to obtain a representative color of the frame.

[0056] When the electronic device **100B** is not connected to the external screen, the processing unit processes a lower area of the frame of the second body **120b** to obtain the representative color of the frame.

[0057] Since the way in which the electronic device **100B** obtains the representative color of the frame is related to whether it is connected to the external screen, the color of the light is changed to allow the user or others to know a screen connection state of the electronic device **100B**.

[0058] FIG. 4A is a schematic partial side view of a closing mode of an electronic device according to another embodiment of the disclosure. FIG. 4B is a schematic partial side view of an unfolding mode of the electronic device in FIG. 4A. FIG. 4C is a schematic perspective view of the unfolding mode of the electronic device in FIG. 4B. FIG. 4D is a schematic view of an illumination direction of a light-emitting module of the electronic device in FIG. 4A.

[0059] Referring to FIGS. 4A to 4D, there are differences between an electronic device **100C** in this embodiment and the electronic device **100** shown in FIG. 1D. The difference is that a rotating module **130c** of the electronic device **100C** is a three-section structure and includes a positioning portion **131c** and two rotating portions **132c**. The positioning portion **131c** is fixed on the rear end RE of a first body **110c** and extends vertically to the upper surface TS. The two rotating portions **132c** are fixed on a second body **120c** and are spaced apart from each other, and are pivotally connected to two sides of the positioning portion **131c**. At least one light-emitting module **140c** includes two, which are disposed on the two rotating portions **132c** respectively.

[0060] The electronic device **100C** includes a light guiding member **160c**, which is disposed in the first body **110c** and parallel to the upper surface TS. The upper surface TS of the first body **110c** corresponding to a range of the guiding member **160c** is formed by the transparent material.

[0061] Referring to FIGS. 4A and 4D, in the closing mode, the light L of each of the light-emitting modules **140c** is illuminated to the rear end RE of the first body **110c** to propagate to the outside world. The light L will be distributed on the rear end RE of the first body **110c** for visual viewing by the outsiders.

[0062] Referring to FIGS. 4B and 4C, in the unfolding mode, the light L of each of the light-emitting modules **140c** is illuminated to the light guiding member **160c**, and the light guiding member **160c** guides the light L to the upper surface TS of the first body **110c**, thereby increasing the light-emitting range of the front end FE and the two sides SE of the first body **110c**.

[0063] Referring to FIG. 4C, when the electronic device **100C** is connected to the external screen, the processing unit processes a lower area of a frame of the second body **120c** to obtain a representative color of the frame. When the electronic device **100C** is not connected to the external screen, the processing unit processes all of the areas of the frame of the second body **120c** to obtain the representative color of the frame.

[0064] Since the way in which the electronic device **100C** obtains the representative color of the frame of the second body **120c** is related to whether it is connected to the external screen, the color of the light is changed to allow the user or others to know a screen connection state of the electronic device **100C**.

[0065] Based on the above, in the electronic device of the disclosure, the light-emitting module is installed on the rotating module, so the illumination angle may be changed with the rotation of the

rotating module. In the closing mode, the light-emitting module faces the rear end of the first body, allowing the light to propagate in a direction away from the user and display it to others. In the unfolding mode, the light-emitting module faces the upper surface of the first body, allowing the light to propagate to the front and the two sides of the first body for display to the user. [0066] Furthermore, the electronic device of the disclosure does not require multiple light-emitting modules to be installed at multiple positions of the electronic device. Compared to the conventional technology, it may improve disadvantages of taking up the space of the body and being expensive.

Claims

1. An electronic device, comprising: a first body; a second body; a rotating module connected to the first body and the second body; and at least one light-emitting module disposed on the rotating module; wherein the second body is adapted to drive the rotating module to rotate relative to the first body along an axial direction to switch to an unfolding mode or a closing mode, in the closing mode, a light-emitting direction of the at least one light-emitting module faces a rear end of the first body, in the unfolding mode, the light-emitting direction of the at least one light-emitting module faces a front end of the first body relative to the rear end.
2. The electronic device according to claim 1, further comprising a bar-shape cover, disposed on the rear end of the first body and has a recess portion and an extension portion, wherein the recess portion accommodates the rotating module and the at least one light-emitting module, and the extension portion is connected to the recess portion and is away from the rear end of the first body.
3. The electronic device according to claim 2, further comprising a reflective member, disposed in the first body and aligned with the recess portion and the extension portion.
4. The electronic device according to claim 2, wherein in the closing mode, a light of the at least one light-emitting module passes through the recess portion to be illuminated to a heat dissipation frame located on the rear end.
5. The electronic device according to claim 3, wherein in the unfolding mode, a light of the at least one light-emitting module passes through the recess portion to be illuminated to the reflective member, and the reflective member reflects the light to the extension portion.
6. The electronic device according to claim 1, wherein the light-emitting direction of the at least one light-emitting module is away from an axial direction of the rotating module.
7. The electronic device according to claim 6, wherein an included angle is between the light-emitting direction and the second body, and the included angle is between 0 degrees and 60 degrees.
8. The electronic device according to claim 2, wherein the rotating module is a three-section structure and comprises two positioning portions and one rotating portion, and the at least one light-emitting module is disposed on the rotating portion.
9. The electronic device according to any one of claim 2 further comprising a frame sensor, disposed on the bar-shape cover or an upper surface and detects a frame of the second body in the unfolding mode.
10. The electronic device according to claim 2, further comprising a light guiding member, disposed in the first body and aligned with the extension portion.
11. The electronic device according to claim 10, wherein the bar-shape cover is formed by a transparent material or a translucent material.
12. The electronic device according to claim 10, wherein in the closing mode, a light of the at least one light-emitting module passes through the recess portion to be illuminated to a heat dissipation frame located on the rear end.
13. The electronic device according to claim 10, wherein in the unfolding mode, a light of the at least one light-emitting module passes through the extension portion to be illuminated to the light guiding member, and the light guiding member guides the light to the extension portion.

- 14.** The electronic device according to claim 10, wherein the rotating module is a five-section structure and comprises two positioning portions and three rotating portions, and the at least one light-emitting module is disposed on at least one of the three rotating portions.
- 15.** The electronic device according to claim 1, further comprising a light guiding member, disposed in the first body.
- 16.** The electronic device according to claim 15, wherein in the closing mode, a light of the at least one light-emitting module is illuminated to the rear end of the first body.
- 17.** The electronic device according to claim 15, wherein in the unfolding mode, a light of the at least one light-emitting module is illuminated to the light guiding member, and the light guiding member guides the light to an upper surface of the first body.
- 18.** The electronic device according to claim 15, wherein the rotating module is a three-section structure and comprises one positioning portion and two rotating portions, and the at least one light-emitting module is disposed on at least one of the two rotating portions.
- 19.** A control method of an electronic device, comprising: the electronic device according to claim 1; and a processing unit coupled to the at least one light-emitting module and controlling the at least one light-emitting module to generate a light.
- 20.** The control method of the electronic device according to claim 19, comprising: a detection unit coupled to the processing unit and configured to detect whether the electronic device is connected to an external screen; wherein if yes, the processing unit controls the external screen to display a frame; and if not, the processing unit controls the second body to display the frame.
- 21.** The control method of the electronic device according to claim 20, comprising: processing, by the processing unit, some of areas or all of the areas of the frame to obtain a representative color of some of the areas or all of the areas; and controlling, by the processing unit, the at least one light-emitting module to generate the light corresponding to the representative color according to the representative color.
- 22.** The control method of the electronic device according to claim 21, comprising: a frame sensor coupled to the processing unit, wherein the frame sensor is adapted to sense a color of an ambient light source; and the processing unit corrects the representative color according to the color of the ambient light source.
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